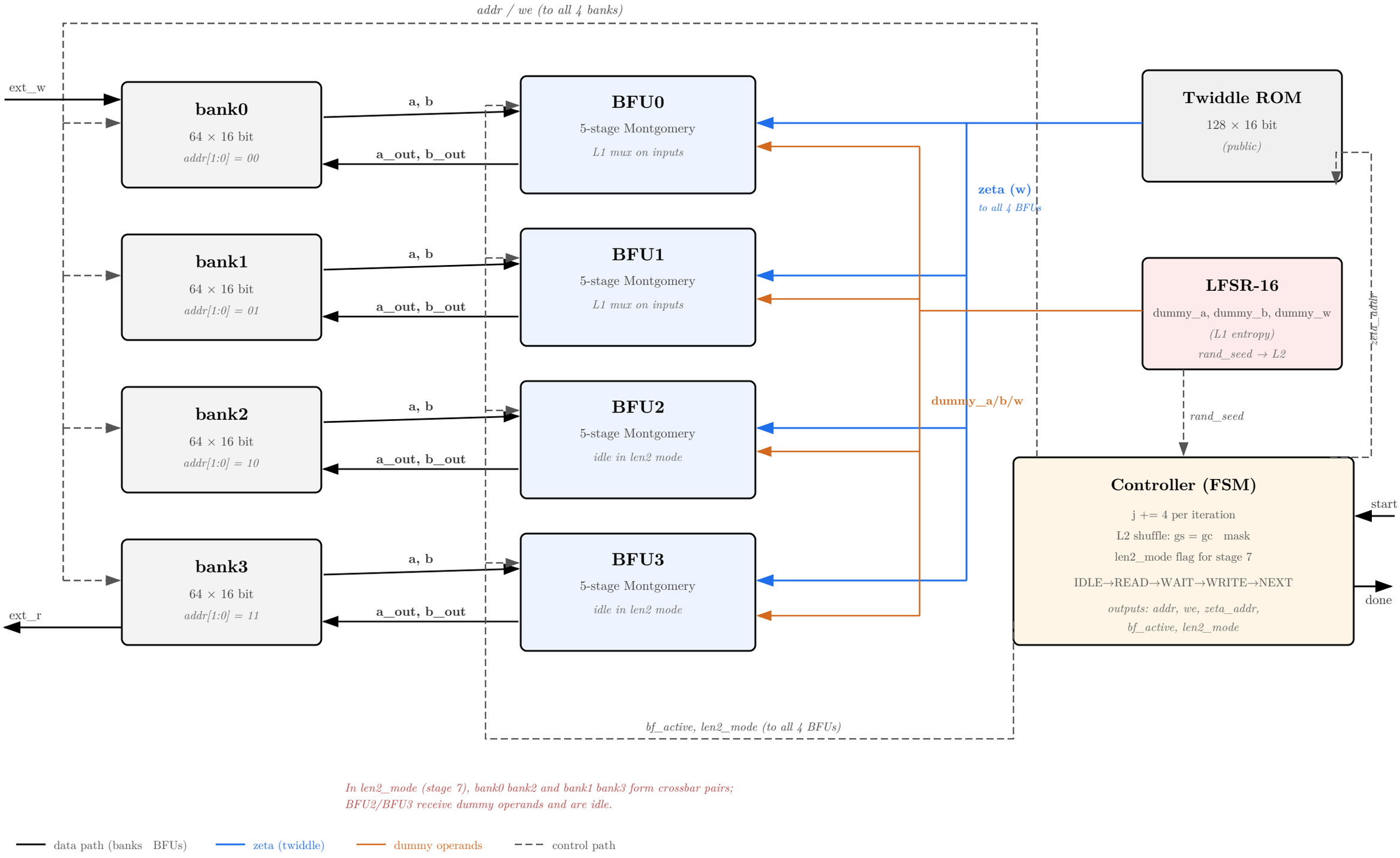

addr / we (to all 4 banks)
ext_w
bank0
64 × 16 bit
addr[1:0] = 00
bank1
64 × 16 bit
addr[1:0] = 01
bank2
64 × 16 bit
addr[1:0] = 10
bank3
64 × 16 bit
addr[1:0] = 11
ext_r
a, b
a_out, b_out
BFU0
5-stage Montgomery
L1 mux on inputs
BFU1
5-stage Montgomery
L1 mux on inputs
BFU2
5-stage Montgomery
idle in len2 mode
BFU3
5-stage Montgomery
idle in len2 mode
Twiddle ROM
128 × 16 bit
(public)
zeta (w)
to all 4 BFUs
dummy_a/b/w
LFSR-16
dummy_a, dummy_b, dummy_w
(L1 entropy)
rand_seed → L2
rand_seed
zeta_addr
Controller (FSM)
j += 4 per iteration
L2 shuffle: gs = gc mask
len2_mode flag for stage 7
IDLE→READ→WAIT→WRITE→NEXT
outputs: addr, we, zeta_addr,
bf_active, len2_mode
start
done
bf_active, len2_mode (to all 4 BFUs)
In len2_mode (stage 7), bank0 bank2 and bank1 bank3 form crossbar pairs;
BFU2/BFU3 receive dummy operands and are idle.
data path (banks BFUs)
zeta (twiddle)
dummy operands
control path